

Notas e Listas de Prob 2

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August 17, 2026

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0 Informações importantes

Devo guardar neste arquivo notas de aula e listas do curso. Sobre as notas, sua maior parte será somente uma enunciação dos principais teoremas vistos, provas interessantes podem ser comentadas e rascunhadas. O código fonte desse arquivo pode ser encontrado em <https://github.com/hnrq104/prob2>.

1 Martingales

Definition 1.1. Let $(\Omega, \mathcal{F}, P)$ be a probability space, a martingale is a sequence $X_1, X_2, \dots$ of random variables associated with σ -algebras $\mathcal{F}_1, \mathcal{F}_2, \dots$ that satisfy:

- (i) $X_i \in L^1$ and $X_i \in \mathcal{F}_i$.
- (ii) $\mathcal{F}_i \subset \mathcal{F}_{i+1}$.
- (iii) $\mathbb{E}[X_{i+1} | \mathcal{F}_i] = X_i$.

Remark 1.2. If one changes the assumption of (iii) to be $\mathbb{E}[X_{i+1} | \mathcal{F}] \leq X_i$ (resp. $\geq$), we call such sequence a supermartingale (resp. submartingale).

Theorem 1.3. (Doob's inequality) Suppose $\{X_j\}$ is a martingale. Then

$$\mathbb{P}(\sup_{j \leq n} |X_j| \geq l) \leq \frac{1}{l} \int \mathbf{1}_{[\sup_{j \leq n} |X_j| \geq l]} |X_n| dP \leq \frac{1}{l} \mathbb{E}|X_n|.$$

Proof. The proof is similar to Kolmogorov's maximal inequality. For $1 \leq j \leq n$, let E_j be the event $\{X_i < l, i < j\} \cap \{X_j \geq l\}$, that is the event that X_j is the first to surpass l . Note that

$$\mathbb{P}(\sup_{j \leq n} |X_j| \geq l) = \sum_{j \leq n} \mathbb{E}[\mathbf{1}[E_j]] \leq \sum_{j \leq n} \frac{1}{l} \mathbb{E}[\mathbf{1}[E_j] |X_j|] \leq \sum_{j \leq n} \frac{1}{l} \mathbb{E}[\mathbf{1}[E_j] |X_n|] = \mathbb{E}[\mathbf{1}[\sup_{j \leq n} |X_j| \geq l] |X_n|] \quad (1)$$

where the second inequality follows from the fact that $\mathbb{E}[X_n | \mathcal{F}_j] = X_j$ and $|x|$ is a convex function. $\square$

Corollary 1.4. (Doob's p-inequality) Let $\{X_j\}$ be a martingale and $p > 1$. Then if, as before, $S_n = \sup_{j \leq n} |X_j|$, we have

$$\mathbb{E}S_n^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|X_n|^p.$$

Proof. Follows from the previous theorem and an integration by parts. Note that for any bounded positive r.v's Y and X that satisfy $\mathbb{P}(Y \geq l) \leq \frac{1}{l} \mathbb{E}[\mathbf{1}[Y \geq l]X]$ we have

$$\begin{aligned} \int Y^p dP &= \int \int_0^Y p y^{p-1} dy dP \\ &= p \int_0^\infty y^{p-1} \mathbb{P}(Y \geq y) dy \\ &\leq p \int_0^\infty y^{p-1} \frac{1}{y} \mathbb{E}[\mathbf{1}[Y \geq y]X] dy \\ &= p \int_0^\infty y^{p-2} \int \mathbf{1}[Y \geq y] X dP \\ &= p \int X \int_0^Y y^{p-2} dy = \frac{p}{p-1} \int XY^{p-1} \\ &\leq \frac{p}{p-1} (\mathbb{E}X^p)^{1/p} (\mathbb{E}Y^p)^{\frac{p-1}{p}} \end{aligned}$$

So that $\mathbb{E}Y^p \leq (\frac{p}{p-1})^p \mathbb{E}X^p$. We can then consider a truncation argument to prove the result for unbounded positive random variables. $\square$

Remark 1.5. This corollary is very cool on its own, it says that if some X_n in the martingale belongs to L^p , then all previous r.v's belong as well.

Exercise 1.1. Consider the martingale of doubling one's bet. In the space $([0, 1], \mathcal{B}([0, 1]), dm)$, we consider the random variables $X_n \in \sigma([j/2^n, (j+1)/2^n])$

$$X_n(\omega) = \begin{cases} 2^n & \text{if } \omega \in [0, 2^{-n}] \\ 0 & \text{elsewhere} \end{cases}$$

It is easy to see that they define a martingale, all in L^1 , $\mathbb{E}X_n = 1$, but $\mathbb{E} \sup_{j \leq n} X_j = (n+1)/2$. So no alternative to [1.4] exists in L^1 .

Exercise 1.2. If $\{X_n\}$ is a martingale with all differences $Y_n = X_n - X_{n-1}$ in L^2 , show that for $n \neq m$, the differences are orthogonal, that is $\mathbb{E}Y_n Y_m = 0$. Therefore

$$\mathbb{E}X_n^2 = \mathbb{E}X_0^2 + \sum_{j=1}^n \mathbb{E}[Y_j^2].$$

Conclude that if $\sup \text{Var}X_n < \infty$, X_n converges in L^2 to a r.v X .

Proof. It follows from the properties of condition expectation. Suppose $n > m$, note that

$$\begin{aligned} \mathbb{E}[(X_n - X_{n-1})(X_m - X_{m-1})] &= \mathbb{E}\mathbb{E}[(X_n - X_{n-1})(X_m - X_{m-1})|\mathcal{F}_{n-1}] \\ &= \mathbb{E}[\mathbb{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}](X_m - X_{m-1})] \\ &= \mathbb{E}[(X_{n-1} - X_{n-1})(X_m - X_{m-1})] = 0 \end{aligned}$$

So the first result follows from that and the cancellation of cross terms. The conclusion holds as $\sum_n \mathbb{E}Y_n^2$ is bounded, so $\{X_n\}$ is a cauchy seq. in L^2 . $\square$

From now on, we denote as $\mathcal{F}$ the smallest σ -field containing $\{\mathcal{F}_n\}$. If $X \in \mathcal{F}$, then $X_n = \mathbb{E}[X|\mathcal{F}_n]$ is a (Doob) martingale, and, by Jensen's ineq, if for some $p \geq 1$ $X \in L^p$ then $X_n \in L^p$ for all n .

Theorem 1.6. If $p \geq 1$ and $X \in L^p$, then $\lim_{n \rightarrow \infty} \|X - X_n\|_p = 0$.

Proof. By the last exercise, the result is true if X is bounded. If not, approximate X by a bounded X' and note that if $\|X - X'\| < \varepsilon$, then $\|\mathbb{E}[X_n|\mathcal{F}_n] - \mathbb{E}[X'|\mathcal{F}_n]\|_p \leq \varepsilon$. So we can do a two step approximation. $\square$

Theorem 1.7. If, for some $p > 1$, a martingale $\{X_n\}$ is bounded in L^p , that is $\sup_n \|X_n\|_p < \infty$, then there exists $X \in L^p$ s.t. $X_n = \mathbb{E}[X|\mathcal{F}_n]$ for all n . In particular $\|X_n - X\|_p \rightarrow 0$.

Proof. As $p > 1$, we can choose a subseq X_{n_j} that weakly converges to a limit X in L^p (just like Egorov's). By noting that for any $A \in \mathcal{F}_m$ we have $\mathbf{1}[A] \in L^q$, we see

$$\mathbb{E}[\mathbf{1}[A]X_m] = \lim_{j \rightarrow \infty} \mathbb{E}[\mathbf{1}[A]X_{n_j}] = \mathbb{E}[\mathbf{1}[A]X].$$

So $X_m = \mathbb{E}[X|\mathcal{F}_m]$. $\square$

Theorem 1.8. Let $X \in L^p$ for $p \geq 1$. Then the martingale $\{X_n\}$ with $X_n = \mathbb{E}[X|\mathcal{F}_n]$ converges to X for almost all ω with respect to P .

Proof. It suffices to prove for $p = 1$. This proof is a bit nuts, we consider the set $\mathcal{M} \subset L^1$ of random variables Y s.t $Y_n \rightarrow Y$ a.e. We show that this set is dense and closed. It is easy to see that it is dense as all variables belonging to $\mathcal{F}_n$ are in it. To see how the set is closed, pick a sequence $Y_j \rightarrow X$ in L^1 . Note that, as $X \in L^1$, by [1.4], X_n is an almost surely bounded sequence so it

suffices to show $\mathbb{P}(\limsup X_n - \liminf X_n > \varepsilon) = 0$ a.e. Set $Y_{j,n} = \mathbb{E}[Y_j | \mathcal{F}_n]$, and note that, for every j

$$\limsup_n X_n - \liminf_n X_n \leq \limsup_n Y_{j,n} - \liminf_n Y_{j,n} + 2 \sup_n |X_n - Y_{j,n}| = 2 \sup_n |X_n - Y_{j,n}|.$$

So

$$\mathbb{P}(\limsup_n X_n - \liminf_n X_n > \varepsilon) \leq \mathbb{P}(\sup_n |X_n - Y_{j,n}| > \varepsilon) \leq \mathbb{P}(|X - Y_j| > \varepsilon) \leq \frac{1}{\varepsilon} \mathbb{E}[|X - Y_j|] \rightarrow 0$$

when j tends to infinity. □

2 Listas

A Resultados clássicos de Convergência